Etiology and Pathogenesis

The scabies mite has four pairs of legs and measures $0.3\mathrm{mm}$ in diameter. It is therefore not easily seen with the naked eye.

The mite cannot fly or jump; it lives its entire 30-day life cycle in and on the epidermis. The female mite burrows into the stratum corneum within 20 minutes and lays approximately 3 eggs per day. The eggs hatch after 4 days, and the larvae migrate to the skin surface, where they mature into adults. After 2 weeks, female and male mites copulate, after which the gravid female burrows back into the stratum corneum. The male, which is slightly smaller than the female, falls off the skin and perishes. The average number of mites a host harbors is usually less than 20, except in **crusted scabies** (previously known as Norwegian scabies), in which a host may harbor over a million mites. Individuals with **HIV**, the elderly, and those with medication-induced immunosuppression are at increased risk of developing crusted scabies, although cases have also been reported in immunocompetent individuals, particularly among indigenous Australians.

Clinical Findings

After initial exposure to the scabies mite, pruritus and rash may take 6 to 8 weeks to develop. With subsequent exposure, symptoms typically appear within a few days due to prior sensitization to mite antigens. The itching is severe and often worse at night. Lesions present as red, scaly, and sometimes crusted (excoriated) papules and nodules, commonly affecting the:

- Interdigital webs
- Sides of the fingers
- Volar aspects of the wrists
- Lateral palms

- Elbows
- Axillae
- Scrotum
- Penis
- Labia
- Areolae in women

A diffuse erythematous eruption on the trunk may occur as a hypersensitivity reaction to mite antigens.

The **pathognomonic lesion** is a **burrow**—a thin, thread-like, linear structure 1 to $10\mathrm{mm}$ in length—representing a tunnel formed by the mite in the stratum corneum. Burrows are most easily seen in the interdigital webs, wrists, or elbows but may be difficult to detect in early disease or after extensive scratching. To enhance visualization, a black felt-tip marker can be rubbed over the area; after wiping with an alcohol pad, residual ink in the burrow makes it appear darker than surrounding skin.

In infants younger than 2 years, infestation may also involve the **face and scalp**, which are typically spared in older children and adults.

Scabies is transmitted primarily through prolonged close personal contact. However, fomite transmission is possible, as the mite can survive off human skin for up to 3 days.

Treatment

Topical scabicides are the preferred treatment. Oral **ivermectin** is an effective alternative, particularly in crusted scabies or when topical therapy is impractical.